

- Q.27 Discuss the application of electrophoresis.
 Q.28 Explain the principle and procedure of PAGE.
 Q.29 Explain the calculation and significance of R_f value in chromatography.
 Q.30 Explain the principle and procedure of HPLC.
 Q.31 Write a note on automation in biochemistry.
 Q.32 Describe the hormones commonly measured in thyroid function tests.
 Q.33 Write a detailed note on glomerular filtration rate (GFR) and factors affecting it.
 Q.34 Discuss the principle and procedure of occult blood estimation in urine analysis.
 Q.35 Describe the normal composition of urine.

SECTION-D

- Note:** Long answer type questions. Attempt any two questions out of three questions. (2x10=20)
 Q.36 Define urine analysis. Explain the urine glucose estimation method with its principle, procedure and clinical significance.
 Q.37 Explain renal function test with its clinical significance.
 Q.38 Explain thyroid function test with its clinical significance.

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4th Sem. / DMLT

Subject : Clinical Biochemistry-IV

Time : 3Hrs.

M.M. : 100

SECTION-A

Note: Multiple choice questions. All questions are compulsory (10x1=10)

- Q.1 The normal color of urine is due to
 a) Urobilin b) Bilirubin
 c) Hemoglobin d) Creatinine
 Q.2 Protein in urine is detected by:
 a) Benedict's test
 b) Biuret test or Sulfosalicylic acid test
 c) Fehling's test
 d) Barfoed's test
 Q.3 Which of the following is a marker of glomerular function?
 a) ALT b) Creatinine
 c) Bilirubin d) Amylase
 Q.4 Steatorrhea refers to
 a) Excess protein in stool
 b) Excess fat in stool
 c) Excess carbohydrate in stool
 d) Excess bile in stool

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Q.5 Electrophoresis is a technique used to separate molecules based on

- a) Size only b) Charge and size
- c) Shape only d) Color only

Q.6 In paper chromatography, the stationary phase is

- a) Liquid
- b) Solid or adsorbent paper
- c) Gas
- d) Buffer solution

Q.7 Stool pH is normally

- a) Acidic (5.5-7.0) b) Neutral (7.0-8.0)
- c) Alkaline (>8.0) d) Very acidic (<4.0)

Q.8 The main purpose of automation in clinical biochemistry is

- a) Reduce accuracy
- b) Increase speed, precision, and reproducibility of tests
- c) Increase manual labor
- d) Reduce patient safety

Q.9 Hyperthyroidism is typically associated with

- a) High TSH, low T4 b) Low TSH, high T4
- c) Low TSH, low T4 d) High TSH, high T4

Q.10 The main hormones measured in thyroid function tests are

- a) T3, T4 and TSH b) Insulin and glucagon
- c) Cortisol and ACTH d) FSH and LH

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SECTION-B

Note: Objective type questions. All questions are compulsory. (10x1=10)

Q.11 Specific gravity of normal urine ranges between _____.

Q.12 What does hematuria indicate?

Q.13 GFR stands for _____.

Q.14 Normal BUN range is .

Q.15 Occult blood in stool indicates.

Q.16 What is RF?

Q.17 SDS-PAGE stands for _____.

Q.18 TLC stands for

Q.19 What is automation?

Q.20 What is meant by TFT?

SECTION-C

Note: Short answer type questions. Attempt any twelve questions out of fifteen questions. (12x5=60)

Q.21 Write the note on urine analysis.

Q.22 Write a short note on GFR.

Q.23 Explain the BUN test.

Q.24 Write the clinical significance of TSH.

Q.25 Write a note on collection and processing of urine sample.

Q.26 Write down the clinical significance of examination of stool sample.

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